

Labwork 8.2 - String Operations

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork8.c**.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (25p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to Labwork 8 String Operations! By using this machine you can do:
1. Encode and Decode
2. Concatenate
3. Count Words
4. Exit
Make your choice:
```

Question 2 - Encode and Decode (25p)

- Create a new function (**en_dec**) that takes a pointer that points the initial character of the string and size of it. (Both the string and the size comes from the main - define the given string below and find out its size by using `strlen()`)
String: aaabbbbcdd
- It will count each letter and return it into this:
a3b4c1d2
- Then it will take this string and return it into its original form:
aaabbbbcdd

```
a3b4c1d2  
aaabbbbcdd
```

Question 3 - Concatenate Two Strings (25p)

- Create a new function (**concat**) that takes two pointers that shows the initial character of each string. Then create a new string with their concatenated version. In addition, convert each space character to _.

String1: a mischievous squirrel cleverly

String2: hid its biggest acorn

- In the end, print the concatenated version of the strings.

```
a_mischievous_squirrel_cleverly_hid_its_biggest_acorn
```

Question 4 - Count Words (25p)

- Create a new function (**count_words**) that takes the pointer that shows the initial character of the string and size of it. (Both the string and the size comes from the main - define the given string below and find out its size by using strlen())
String: The mountains are calling and I must go
- This function will count the words, return it and print each word separately.

```
The  
mountains  
are  
calling  
and  
I  
must  
go
```

```
In this sentence there are 8 words.
```